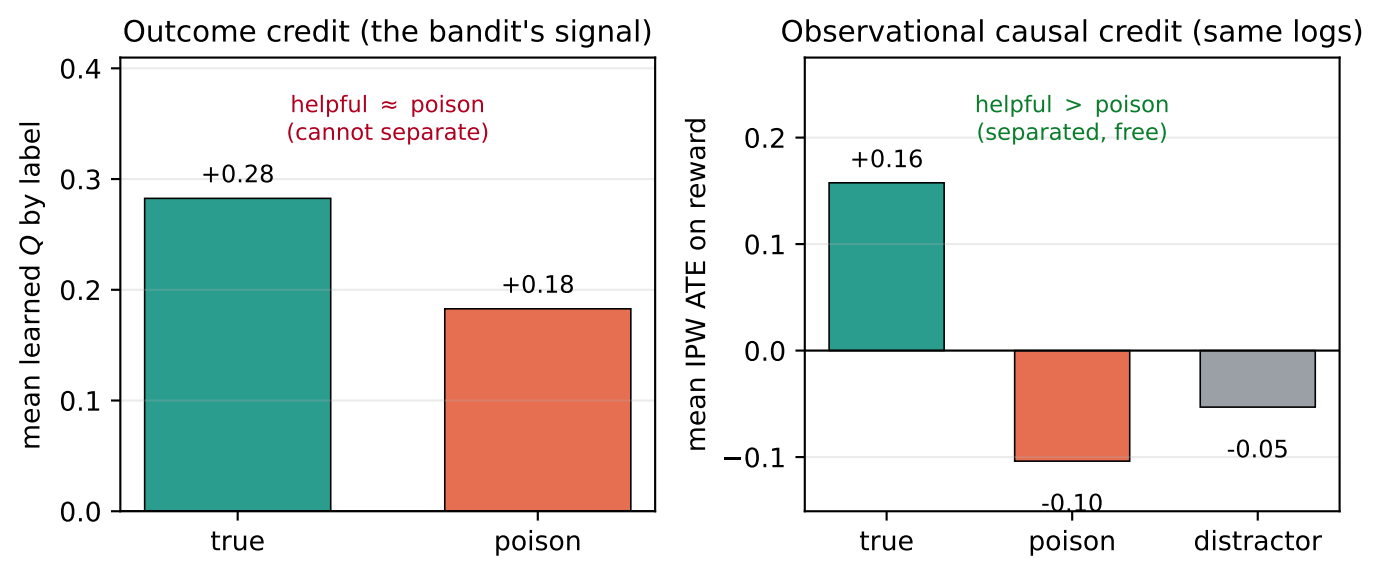
Outcome credit (the bandit's signal)
mean learned Q by label
0.4
0.3
0.2
0.1
0.0
helpful ≈ poison
(cannot separate)
+0.28
+0.18
true
poison
Observational causal credit (same logs)
mean IPW ATE on reward
0.2
0.1
0.0
−0.1
helpful > poison
(separated, free)
+0.16
-0.10
-0.05
true
poison
distractor